

Proof: Let A be an $n \times n$ invertible matrix. By definition, there exists an $n \times n$ matrix A^{-1} s.t. $AA^{-1} = A^{-1}A = I_n$.

Book's proof

Suppose that A is invertible. Then since the equation $Ax=b$ has a solution for each b (Theorem 5), A has a pivot position in every row (Theorem 4 in Section 1.4). Because A is square, the n pivot positions must be on the diagonal, which implies that the reduced echelon form of A is I_n . That is, $A \sim I_n$.

Now suppose, conversely, that $A \sim I_n$. Then, since each step of the row reduction of A corresponds to the left-multiplication by an elementary matrix, there exist elementary matrices $E_1, \dots, E_p$ such that

$$A \sim E_1 A \sim E_2 (E_1 A) \sim \dots \sim E_p (E_{p-1} \dots E_1 A) = I_n$$

That is, $\boxed{E_p \dots E_1 A = I_n} \quad (1)$

Since the product $E_p \dots E_1$ of invertible matrices is invertible, (1) leads to $(E_p \dots E_1)^{-1} (E_p \dots E_1) A = (E_p \dots E_1)^{-1} I_n$

$$A = (E_p \dots E_1)^{-1}$$

Thus A is invertible, as it is the inverse of an invertible matrix (Theorem 6). Also, $A^{-1} = [(E_p \dots E_1)^{-1}]^{-1} = E_p \dots E_1$

Then $A^{-1} = E_p \dots E_1 \cdot I_n$, which says that A^{-1} results from applying $E_1, \dots, E_p$ successively to I_n . This is the same sequence in (1) that reduced A to I_n .

□

An Algorithm for Finding A^{-1}

If we place A and I side by side to form an augmented matrix $[A \ I]$, then row operations on this matrix produce identical operations on A and on I . By Theorem 7, either there are row operations that transform A to I_n and I to A^{-1} or else A is not invertible.